Staphylococcal Scarlet Fever

The exanthem of staphylococcal scarlet fever tends to be more tender than that seen in streptococcal scarlet fever. Systemic signs such as malaise and fever are invariably present. Within a few days of rash onset, thick desquamation begins, and the entire skin may peel over the following week.

Unlike generalized staphylococcal scalded skin syndrome (SSSS), the scarlatina-form eruption is not associated with bullae or superficial epidermal exfoliation and can be clinically challenging to distinguish from other infectious erythrodermal conditions such as toxic shock syndrome (TSS) and streptococcal scarlet fever. A key distinguishing feature of staphylococcal scarlet fever is the absence of pharyngitis, which is typically present in streptococcal-mediated disease. Instead, the infection usually originates from a cutaneous focus.

A study by Wang and colleagues evaluated 20 children with staphylococcal scarlet fever. All infections originated from the skin: 16 cases from furuncles or carbuncles, and 2 each from abscesses or wound infections. All *Staphylococcus aureus* isolates expressed staphylococcal enterotoxin B (SEB). Notably, SEB shares significant protein sequence homology with streptococcal pyrogenic exotoxin A (SPEA), a toxin linked to streptococcal scarlet fever. However, other studies have implicated additional staphylococcal enterotoxins in similar cases. This variability in toxin profiles may reflect the clinical basis of diagnosis. It has been proposed that staphylococcal scarlet fever may represent an incomplete form of TSS, where toxins spread from a skin source, preferentially activating skin-associated lymphoid tissue rather than mucosal-associated lymphoid tissue.

Given that streptococcal scarlet fever typically follows pharyngitis, the presence of a scarlet fever–like rash without pharyngitis should prompt evaluation for a localized skin infection (e.g., furuncle) that can be cultured to confirm the diagnosis.

Differential Diagnosis

The diagnosis of scarlet fever is primarily clinical and supported by positive bacterial cultures. The differential diagnosis includes other toxin-mediated conditions such as SSSS.

- **Kawasaki disease** shares features such as mucosal changes (e.g., strawberry tongue), extremity swelling, and palmoplantar desquamation during recovery. However, Kawasaki disease is characterized by prolonged fever and negative bacterial cultures.
- **Atypical drug hypersensitivity reactions** may mimic some cutaneous findings but lack mucosal involvement. These reactions are typically associated with a history of recent drug exposure and peripheral eosinophilia.
- **Streptococcal vs. staphylococcal scarlet fever**: Streptococcal cases usually arise from pharyngitis, whereas staphylococcal cases originate from a skin infection.

Treatment and Prognosis

Treatment involves antibiotics to eliminate the causative organism. If a localized skin infection (e.g., abscess, furuncle, or carbuncle) is present, incision and drainage are essential. Unlike streptococcal scarlet fever, staphylococcal scarlet fever is not associated with acute rheumatic fever or post-streptococcal glomerulonephritis. In cases with systemic features resembling TSS, management should follow TSS treatment guidelines.

Recurrent Toxin-Mediated Perineal Erythema

In 1996, Manders and colleagues described a novel condition termed **recurrent toxin-mediated perineal erythema**. This disorder is characterized by a sudden onset of striking, diffuse macular erythema in the perineal area, appearing 24 to 48 hours after pharyngitis caused by toxin-producing group A *Streptococcus* or *S. aureus*.

Clinical features often overlap with scarlet fever and may include: - Strawberry tongue - Perineal erythema and edema - Subsequent palmoplantar desquamation

Notably, systemic signs such as high fever, hypotension, or other manifestations of TSS or scarlet fever are typically absent. Diarrhea is a common accompanying symptom. Recurrences are more frequent in this localized variant compared to classic scarlet fever.